



SAMPLE ROOM PLANNING REQUIREMENT

Group 02



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GROUP 02

Registration Number

Member Name

S / 18 / 812

J.M.K.K.Jayasinghe

S / 18 / 813

K.M.G.T.Kariyawasam

S / 18 / 814

R.M.N.Kaushalya

S / 18 / 815

H.M.U.M.Keerthisinghe

S / 18 / 816

S.Kumaran

S / 18 / 817

M.V.T.Magammana

S / 18 / 818

F.B.D.P.Mahindarathna

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ABSTRACT

Assigning orders to modules or sewing lines plays a major role in the garment industry whether it is bulk orders or sample orders. The objective of our project is to optimize the assignment of sample orders to modules in the sample room. Our project encompasses the sample room tracking system, sample room planning, and sample room data entry. We utilize transportation algorithm to formulate and solve the problem. We implement it in the coding process with the assistance of *lp solve* package in r programming language. Also we create a user interface with the help of Shiny package in r. We use sample past sewing order dataset and module details dataset as our data sources. The result of the code is presented using a table indicating which order is assigned to which module. We face several challenges in the implementation process the main challenge was choosing an algorithm to apply for this scenario. We also tried using critical path method, job sequencing, genetic algorithms, and greedy approaches. We would like to discuss with Brandix group and do further enhancement to the implementation according to their further requirements. This project signifies the effectiveness of using transportation algorithm to optimize assignment of sample orders to different sewing modules in the sample room.

01. INTRODUCTION

01.01 Company Overview



Brandix has been providing renowned brands all over the world with inspired solution for over 40 years as a leading apparel solution provider in Sri Lanka. They in an upward direction coordinated production network, extending worldwide organization of assembling and working areas, an enlivened representative base and foundation have assisted us with conveying a large number of pieces of clothing to their clients. Brandix has established itself as a market leader in the apparel industry thanks to its expertise in product innovation, research and development, unparalleled speed of delivery, and unwavering dedication to delivering an outstanding product.

01.02 What is a Sample?

In the readymade garment industry, sample garments are very important. A sample is a representation of the manufacturing, quality, and output of the complete order for garment exports that everyone can understand. The sample is prepared in the sample room, according to instructions from the buyer. It can guarantee the pre- and post-condition of the requested garment for both the consumer and the buyer of the clothing. Additionally, the sample is utilized to get the necessary market insights regarding the order's business promotions.

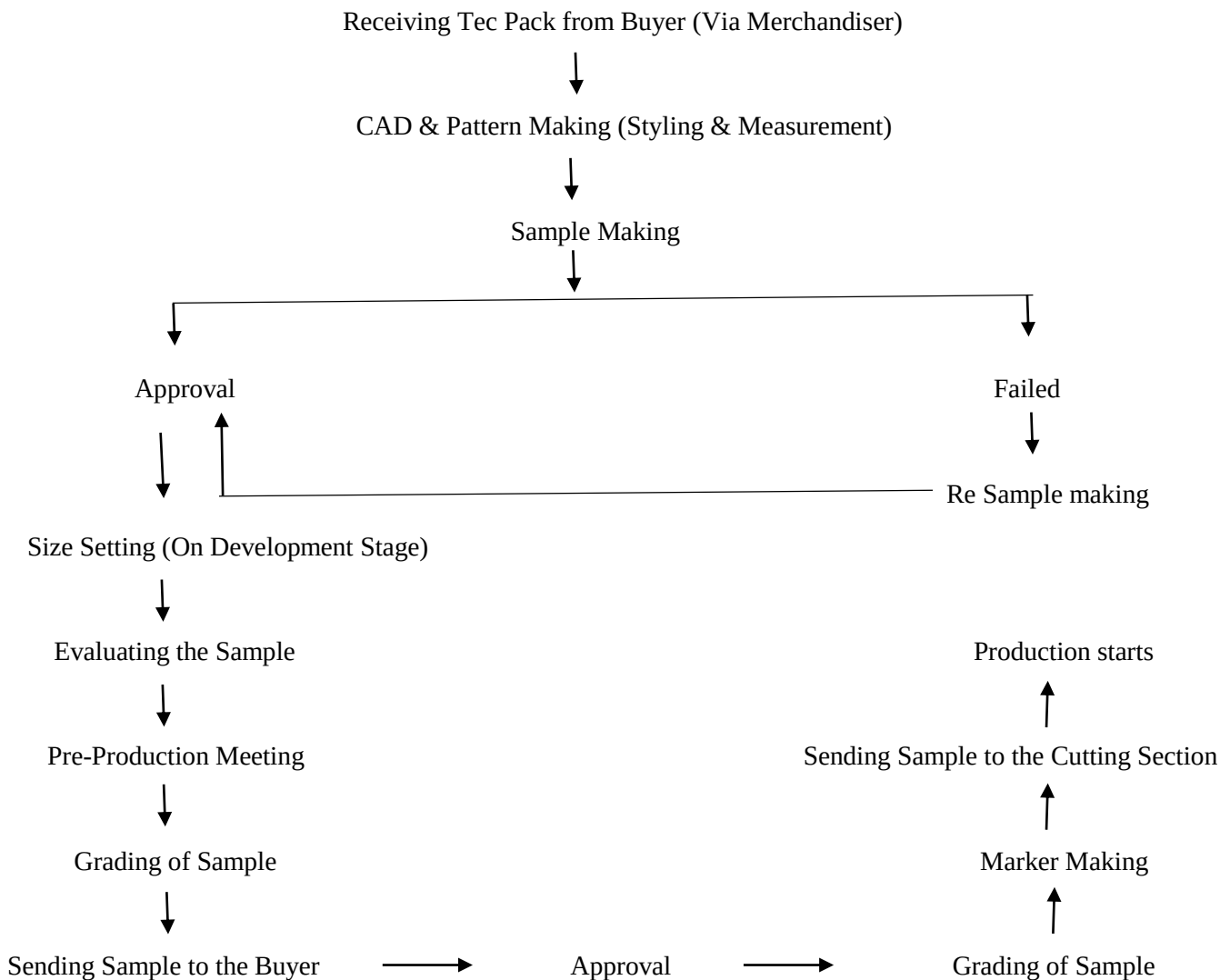


Figure 1.1: Sampling Process Flow Chart

01.03 What is Sample Room?

The sample room is a dedicated space within a garment manufacturing facility in Sri Lanka. That's where concepts for garment designs are taken from drawings to actual clothing. These samples serve as prototypes and references for bulk production.

The most experienced and productive worker was present in the sample room area. Experts in their respective fields, fashion designers, pattern makers, sample cutters, fabric specialists, sample machinists, and fit specialists make up the sample room.

Sample room process can be divided into following steps.

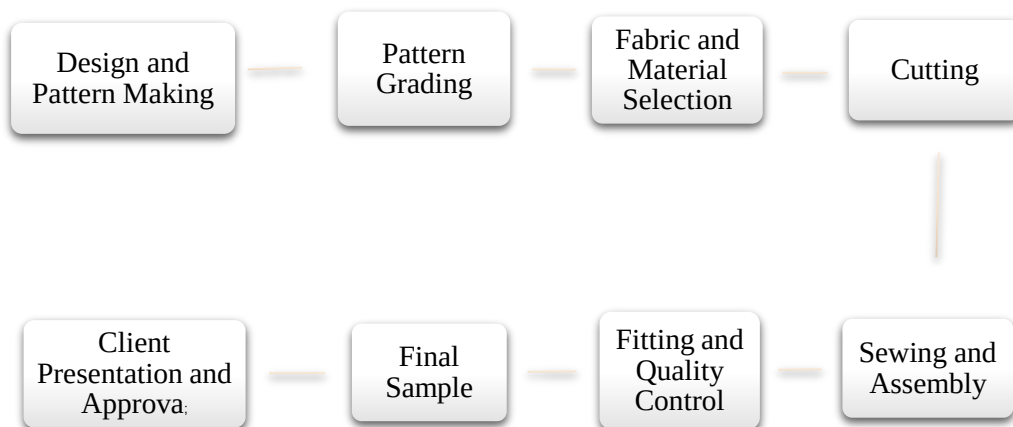


Figure 1.2: Steps of sample room process

01.04 Problem Statement

As previously mentioned, the sewing procedure in the sample room requires the use of multi-skilled machine operators. The production plan is created in the sample room by assigning one machine operator to one garment. That implies a tailor must sew the entire garment until it is complete. There are five modules in the sample room. In a module, there are five skilled persons.

The sample room specifies who will operate the machine for each outfit. They are responsible for their sample and must submit the completed sample on time.

The SMV for sewing the garment should be provided in the sample room production plan. These SMVs for sample sewing differ from the SMVs for bulk production. The SMV for sampling is greater than the SMV for bulk production.

The samples in the manufacturing plan indicate their sample stage. We can determine what samples will be generated on a specific day and what state they are in by looking at the production plan.

The sample room's planning division is in charge of creating the production plan for the sewing and cutting procedures. Making a production plan for the sample room is important since it allows them to manage their sample-producing procedure and deliver their goods to the buyer on time. The sample production plan differs from the bulk production plan in that the sample room only produces for one day. However, the bulk manufacturing plan is for one month.

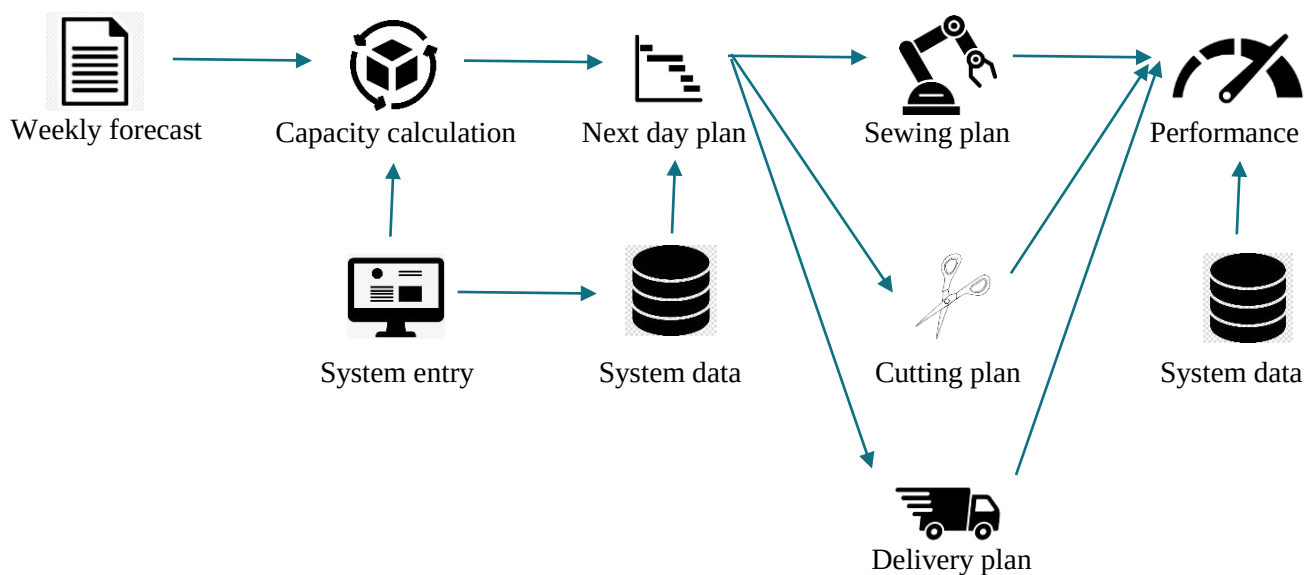


Figure 1.3: Planning Process

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After the merchant enters the details about the order into the SPTS system we want to assign that order to the suitable modular

Key considerations of assigning orders to sample room modules

1. Order Prioritization

- Determine the priority of each order based on factors such as customer requirements, delivery deadlines, and production volume. High-priority orders should be assigned first.

2. Module Specialization

- Divide the sample room into specialized modules or workstations, each focusing on specific tasks, such as pattern development, cutting, sewing, and quality control.
- Assign orders to modules based on the nature of the work involved. For instance, orders requiring complex patterns may be assigned to the pattern development module.

3. Workload Distribution

- Balance the workload across modules to ensure that no module is overloaded while others remain underutilized.
- Consider the skill sets of the workers in each module and match them with the requirements of the orders.

01.05 Project Scope

- **Sample room tracking system:**

We want to track the system and get the details about the processing orders and which modules they are assigned to.

- **Sample room planning:**

After tracking the system and getting an idea about which modules are free or when modules are free. Then look at the above conditions if all these are satisfied then we assign the order to the suitable module.

- **Sample room data entry:**

Enter the data about the order into the system. Enter the style and time duration to finish that order by considering past orders' time duration details.

Since our problem statement was with the sample room concept, we just took the sewing details into consideration here.

We gathered those data from the overall collection of data. Certain styles' stitching times vary slightly with other, which we believe that they might be a new pattern and those values consider as outlier values. Module capacity & stitching times are more useful for get the optimal results among these datasets. We created an objective function and constraints using this data, and then we arrived at the best solution.

We take a linear programming approach to solve this problem and implement it using two R packages *lpSolve*, *Shiny*. We use past sewing order details, module details datasets as our data sources. And represent the assignment using a table indicating which order is assigned to which module.

01.06 Expected Benefits (Quantifiable)

Eliminate manual data entering

Reduce planning effort by 50%

01.07 Report Organization

The report is organized into six chapters.

- 01 Introduction
- 02. Literature Review – reviewing existing methodologies
- 03. Methodology - use transportation algorithm to formulate and solve a problem.
- 04. Data Analysis – used data set for solve problem
- 05. Result and Discussion – analyze the result
- 06. Conclusion

02. LITERATURE REVIEW

02.01 Current state

Transportation algorithms play a crucial role in optimizing resource allocation and production scheduling within manufacturing systems. Efficient transportation planning ensures that goods are moved from suppliers to production facilities and then to distribution centres or customers in a cost-effective and timely manner. This literature review explores key concepts, methodologies, and recent advancements in transportation algorithms and production scheduling.

02.02 Related works

Currently, lot of python code have use to solve line assignment problems.

1. Assignment Problem

Link : <https://machinelearninggeek.com>

A problem that requires pairing two sets of items given a set of paired costs or profit in such a way that the total cost of the pairings is minimized or maximized. The assignment problem is a special case of linear programming.

- **Solving step**

- Step 01: Problem Formulation

- Step 02: Initialize LP Model

- Step 03: Define Decision Variable

- Step 04: Define Objective Function

- Step 05: Define the Constraints

- Step 05: Solve Model

In this article, the authors have written about solving linear programming problems using python *PuLp* library. They have solved Assignment problem using this library. This is a simple case study but they have mentioned the possibility of extending this to complex case studies like Cargo Loading problems, Staff scheduling problems.

2. ASSIGNMENT OF MULTIPLE JOBS SCHEDULING TO A SINGLE MACHINE

Link:https://www.researchgate.net/publication/349678365_Assignment_of_multiple_jobs_scheduling_to_a_single_machine.

In this paper, the authors have modified the existing Hungarian Method to address the challenge of limiting the number of jobs assigned to a single machine in the unbalanced assignment problem. The algorithm was implemented in Python 3.9.0.

The assumptions of the *multiple assignment* model are provided below:

- (i) The number of jobs is always greater than the number of machines.
- (ii) Each machine can do more than one job.
- (iii) No same job is assigned to more than one machine.
- (iv) Each machine at least performs one job and no job is left without assigning to machine.
- (v) A machine can perform a job one after the other if it is to do more than one job.
- (vi) The total assigning time of each machine is less than or equal to its available time.

They have considered the above assumptions in solving the problem.

In this proposed work, a new and simple method was introduced for assigning different activities on the available resources. In this method, they include one more parameter as time for each resource and it can be replaced by cost for assigning cost-related problems. This method fulfills the requirements of assigning more activities to one resource within a limit and obtaining the optimal result.

02.03 Limitations

- **Capacity Constraints and Resource Optimization:**

In the context of the current project, the consideration of daily capacity constraints on each sewing line is a notable aspect. Research in this area emphasizes the importance of managing and optimizing resources effectively. Techniques such as capacity planning, production levelling, and advanced scheduling algorithms contribute to achieve a balance between demand and production capabilities.

- **Time Complexity and Scalability:**

While transportation algorithms provide powerful solutions, it is essential to consider their time complexity and scalability, especially in large-scale manufacturing environments. Recent advancements focus on developing algorithms that are not only optimal but also computationally efficient, enabling real-time decision-making in dynamic production environments.

03. METHODOLOGY

03.01 Data Collection

Data collection is an essential phase in every type of project. Precise data is a must to ensure the accuracy of the results. We followed the below steps in the data collection phase.

- 1. Sewing Time**

We get a sample data set from Brandix Group on past sample room orders. From that data set, we filter the orders from each garment style and calculate the time taken for the sewing phase by subtracting the sewing end date by the sewing start date.

- 2. Order Data**

From the given data set we identified the set of customers and their requesting styles, range of average quantities they ordered. Then we generate a set of random orders according to the acquired data.

- 3. Sewing Line Capacities**

Brandix Group has shared the details about the modules in the sample room section. Including the number of operators in each module. Working hours of the workers per day, number of modules working.

03.02 Data Preprocessing and Integration

In the above calculated sewing time data, we remove the outliers, divide them by ordered quantity and calculate the time taken to sew one piece of garment.

Then we combine two datasets styles_data and orders_data by taking style type as common. Then we calculate the total time taken to process the orders by multiplying the order quantity by the time taken to sew one garment.

03.03 Construction of Cost Matrix

Constructing a cost matrix is a key step in the transportation algorithm. The cost matrix indicates the costs between each style and each module. We consider the efficiency of each line on each garment style and then assign ranks where the most efficient line for that style has the rank 1. We take those ranks as the cost values.

Table 3.1: Cost Matrix

	Module 1	Module 2	Module 3	Module 4	Module 5
FLARE LEGGING	4	2	3	5	1
CS MODAL RIB HENLEY BODYSUIT	1	4	1	2	3
MESH RACERBACK FLOW DRESS	2	3	5	4	2
Fitted Long Sleeve Crew Neck Mesh Shirt	3	1	4	3	5
Swing Long Sleeve Scoop Neck Tunic	5	5	2	1	4

03.04 Demand and Supply Vector Calculation

Calculating the demand and supply vectors are pivotal part in transportation algorithm. Here we take total time taken to sew garments from the styles as the demand values. Daily capacities of each modules as the supply values.

Demand Vector

28.40 26.96 108.15 13.20 20.00

Supply vector

44.25 44.25 44.25 44.25 44.25

03.05 Linear Programming Formulation

Transportation Algorithm

Let, X_{ij} be the number of hours j^{th} module contribute to i^{th} style where $i = 1, 2, 3, 4, 5$ modules and $j = 1, 2, 3, 4, 5$ for the five styles respectively.

$Z =$ Total Weighted Time

$$\begin{aligned} \text{Min } Z = & 4X_{11} + 2X_{12} + 3X_{13} + 5X_{14} + 1X_{15} + 1X_{21} + 4X_{22} + 1X_{23} + 2X_{24} + 3X_{25} + 2X_{31} + 3X_{32} + \\ & 5X_{33} + 4X_{34} + 2X_{35} + 3X_{41} + 1X_{42} + 4X_{43} + 3X_{44} + 5X_{45} + 5X_{51} + 5X_{52} + 2X_{53} + 1X_{54} + 4X_{55} \end{aligned}$$

Subject to:

Demand Constraint:

$$X_{11} + X_{12} + X_{13} + X_{14} + X_{15} \geq 28.40$$

$$X_{21} + X_{22} + X_{23} + X_{24} + X_{25} \geq 26.96$$

$$X_{31} + X_{32} + X_{33} + X_{34} + X_{35} \geq 108.15$$

$$X_{41} + X_{42} + X_{43} + X_{44} + X_{45} \geq 13.20$$

$$X_{51} + X_{52} + X_{53} + X_{54} + X_{55} \geq 20.00$$

Supply Constraint:

$$X_{11} + X_{21} + X_{31} + X_{41} + X_{51} \leq 44.25$$

$$X_{12} + X_{22} + X_{32} + X_{42} + X_{52} \leq 44.25$$

$$X_{13} + X_{23} + X_{33} + X_{43} + X_{53} \leq 44.25$$

$$X_{14} + X_{24} + X_{34} + X_{44} + X_{54} \leq 44.25$$

$$X_{15} + X_{25} + X_{35} + X_{45} + X_{55} \leq 44.25$$

$$X_{ij} \geq 0 \text{ for } i = 1, 2, 3, 4, 5 \text{ and } j = 1, 2, 3, 4, 5$$

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Table 3.2: Optimal Assignment of Orders to Modules by the Style Type

	Module 1	Module 2	Module 3	Module 4	Module 5
FLARE LEGGING	0	0	0	0	1
CS MODAL RIB HENLEY BODYSUIT	0	0	1	0	0
MESH RACERBACK FLOW DRESS	1	0	0	0	0
Fitted Long Sleeve Crew Neck Mesh Shirt	0	1	0	0	0
Swing Long Sleeve Scoop Neck Tunic	0	0	0	1	0

This binary matrix illustrates the optimal assignment of orders to modules by the style type.

Sensitivity Analysis provides a full analysis of the LP model's results and constraints.

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Microsoft Excel 15.0 Sensitivity Report

Worksheet: [Book1.xlsx]Sheet1

Report Created: 12/3/2023 4:59:03 PM

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$13:\$G\$17						
\$C\$13	FLARE LEGGING Module 1	0	3	4	1E+30	3
\$D\$13	FLARE LEGGING Module 2	0	0	2	1E+30	0
\$E\$13	FLARE LEGGING Module 3	0	0	3	1E+30	0
\$F\$13	FLARE LEGGING Module 4	0	2	5	1E+30	2
\$G\$13	FLARE LEGGING Module 5	28.4	0	1	0	3
\$C\$14	CS MODAL RIB HENLEY BODYSUIT Module 1	0	2	1	1E+30	2
\$D\$14	CS MODAL RIB HENLEY BODYSUIT Module 2	0	4	4	1E+30	4
\$E\$14	CS MODAL RIB HENLEY BODYSUIT Module 3	26.96	0	1	1	1
\$F\$14	CS MODAL RIB HENLEY BODYSUIT Module 4	0	1	2	1E+30	1
\$G\$14	CS MODAL RIB HENLEY BODYSUIT Module 5	0	4	3	1E+30	4
\$C\$15	MESH RACERBACK FLOW DRESS Module 1	44.25	0	2	2	1E+30
\$D\$15	MESH RACERBACK FLOW DRESS Module 2	31.05	0	3	0	1
\$E\$15	MESH RACERBACK FLOW DRESS Module 3	0	1	5	1E+30	1
\$F\$15	MESH RACERBACK FLOW DRESS Module 4	17	0	4	0	1
\$G\$15	MESH RACERBACK FLOW DRESS Module 5	15.85	0	2	2	0
\$C\$16	Fitted Long Sleeve Crew Neck Mesh Shirt Module 1	0	3	3	1E+30	3
\$D\$16	Fitted Long Sleeve Crew Neck Mesh Shirt Module 2	13.2	0	1	1	2
\$E\$16	Fitted Long Sleeve Crew Neck Mesh Shirt Module 3	0	2	4	1E+30	2
\$F\$16	Fitted Long Sleeve Crew Neck Mesh Shirt Module 4	0	1	3	1E+30	1
\$G\$16	Fitted Long Sleeve Crew Neck Mesh Shirt Module 5	0	5	5	1E+30	5
\$C\$17	Swing Long Sleeve Scoop Neck Tunic Module 1	0	6	5	1E+30	6
\$D\$17	Swing Long Sleeve Scoop Neck Tunic Module 2	0	5	5	1E+30	5
\$E\$17	Swing Long Sleeve Scoop Neck Tunic Module 3	0	1	2	1E+30	1
\$F\$17	Swing Long Sleeve Scoop Neck Tunic Module 4	20	0	1	1	1
\$G\$17	Swing Long Sleeve Scoop Neck Tunic Module 5	0	5	4	1E+30	5

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$C\$18:\$G\$18 <= \$C\$20:\$G\$20						
\$C\$18	Module 1	44.25	-2	44.25	17	7.25
\$D\$18	Module 2	44.25	-1	44.25	17	7.25
\$E\$18	Module 3	26.96	0	44.25	1E+30	17.29
\$F\$18	Module 4	37	0	44.25	1E+30	7.25
\$G\$18	Module 5	44.25	-2	44.25	17	7.25
\$H\$13:\$H\$17 >= \$J\$13:\$J\$17						
\$H\$13	FLARE LEGGING	28.4	3	28.4	7.25	17
\$H\$14	CS MODAL RIB HENLEY BODYSUIT	26.96	1	26.96	17.29	26.96
\$H\$15	MESH RACERBACK FLOW DRESS	108.15	4	108.15	7.25	17
\$H\$16	Fitted Long Sleeve Crew Neck Mesh Shirt	13.2	2	13.2	7.25	13.2
\$H\$17	Swing Long Sleeve Scoop Neck Tunic	20	1	20	7.25	20

Figure 3.1: Sensitivity Report

03.06 Software and Tools

We use *lp.transport* and *lp.assign* functions from *lpSolve* package. *Lp_solve* is freely available software for solving linear, integer and mixed integer programs.

```
lp.transport (cost.mat, direction="min", row.signs, row.rhs, col.signs, col.rhs, presolve=0, compute.sens=0,  
integers = 1:(nc*nr) )
```

- *cost.mat* - Matrix of costs
- *direction* - Objective "min" or "max"
- *row.signs* - Vector of character strings giving the direction of the row constraints: each value should be one of "<," "<="," "=", "==" ,">," or ">=".
- *row.rhs* - Vector of numeric values for the right-hand sides of the row constraints.
- *col.signs* - Vector of character strings giving the direction of the column constraints: each value should be one of "<," "<="," "=", "==" ,">," or ">=".
- *col.rhs* - Vector of numeric values for the right-hand sides of the column constraints.

```
lp.assign (cost.mat, direction = "min", presolve = 0, compute.sens = 0)
```

- *cost.mat* - Matrix of costs
- *direction* - Objective "min" or "max"

Also, we use Shiny package to build the user interface. Shiny is an open source R package that provides an elegant and powerful web framework for building web applications using R.

We Microsoft Excel to perform the sensitivity analysis.

03.07 Assumptions

- All 5 modules have the same capacity.
- One order cannot be assigned to several modules.
- All the orders are received at one time.
- The following are the only customers.
 - SKUNK KIDS WEAR
 - VS - Casual Sleep
 - VS – Sport
 - LANE BRYANT
- The following are the styles.
 - Flare Legging
 - Swing Long Sleeve Scoop Neck Tunic
 - Cs Modal Rib Henley Bodysuit
 - Fitted Long Sleeve Crew Neck
 - Mesh Racerback Flow Dress
- The time taken to sew the sample does not change according to sample type (proto, testing, try, photo-shoot) and size (XS, S, M).

03.08 Alternative Approaches

There are few other algorithms that we can use for this problem.

- Integer Linear Programming
- Mixed-Integer Linear Programming
- Metaheuristic Algorithms
- Heuristic Approaches
- Simulation and Monte Carlo Methods
- Queuing Theory
- Constraint Programming
- Multi-Objective Optimization

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- Column Generation
- Dynamic Programming

There are several plus points of choosing transportation algorithm in this scenario. The main advantage is simplicity and efficiency. Due to the linear structure of the algorithm it is easy to implement in software. Also, we can easily interpret the results.

04. DATA RESOURCES

04.01 How we obtained our data?

When we went to the Brandix fast fashion Welisara Center (BFFC), we learned about the workings of the production line. We also spoke with internal supervisors, who provided us with further information regarding the Sample Room Plan. Following that, we went to Brandix Fast Fashion Mirigama 1 (BFFMR), where they showed us the process of the factory and went over everything in detail. Following that, we set up online sessions with them, and during these conversations, they provided us with more information about the sample room process.

Here we are considering only a sample room plan.

04.02 Main Data Sets

1. Sewing data

SEW1 PACKAGE PROCESS START	SEW1 SEWING START	SEW1 SEWING END	SEW1 QC START	SEW1 QC END
2023-03-04 08:56:42	2023-03-07 10:56:12	2023-03-07 17:17:52	3-7-2023 17:17:00	2023-03-08 14:15:53
2023-03-11 09:17:42	2023-03-14 11:58:24	2023-03-14 17:56:55	2023-03-14 17:56:55	2023-03-14 19:14:36
2023-03-16 09:24:42	2023-03-17 08:41:57	2023-03-17 19:08:20	2023-03-17 19:08:20	2023-03-22 13:41:43
2023-03-15 11:45:22	2023-03-15 16:10:54	2023-03-20 11:39:50	2023-03-20 11:39:50	2023-03-21 13:47:12

Figure 4.1: Sewing Data

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In Figure 4.1 shown part of data set of sewing details. That data we collected from overall data set (Brandix provided that through the zoom meetings). There are cutting details, embellishment details, delivery details, Sewing details, etc. But here we considered only Sewing details because, our problem statement regarding Sample Room Plan.

According to the figure 4.1 we can see sew 1 package process start time and sewing start time are different that means these two jobs are doing by separate persons, although one person setup the package process other person by being late sewing start time also be late. That is the reason for the difference between two times. Sometimes there create the new patterns, in such cases occur sewing start and end time has a significant difference. So it becomes an outlier such times are more effective for the optimal solution.

Although, the SRFID is different, the styles are the same, so after grouping them, the average value of the sewing time taken for each style was used for calculation.

- Sew1 Package process start: In here after the cutting part setup the need equipment for sewing.
- Sew 'i' Sewing Start Time & End; where $i = 1, 2, 3$

Sew1 for basic sewing part

Sew 2 & Sew 3 for additional sewing parts

- Sew 'i' QC start & End Time: In this time period check the quality of each sew 'i'

Full CSV file with sewing data set: <https://classroom.google.com/w/NjAwNzU0MjI3MTM1/t/all>

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2. Capacity Details

In the Sample room has five modules and each module has five persons.

In a one module capacity details are here (Table 4.2),

Table 4.2: Capacity Details

Module 1		WH	OT	DT	Available time
	Person 1	8	1.25	1	8.25
	Person 2	8	1.25	0	9.25
	Person 3	8	1.25	0	9.25
	Person 4	8	1.25	0	9.25
	Person 5	8	1.25	1	8.25
					44.25

According to this Table 4.2 we can collect each person,

❖ *WH (working hour)* –

Total hours worked, exclusive of meals and rest. (8 hr)

❖ *OT (Overtime in hours)* –

Overtime refers to any hours worked by an employee that exceed their normally scheduled working hours. While a generalized overtime definition refers simply to those hours worked outside of the standard working schedule, overtime commonly refers concurrently to the employee's remunerations of such work.

❖ *DT (Doing time)* –

If they go for official work like an official meeting or society work or somewhat that time counts as working time

❖ *Available time* –

Available time for working in one day, this time can be calculated using this equation,

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$$\text{Available Time} = (WH + OT - DT) \rightarrow (1)$$

❖ *Total Model Capacity* = Sum of Available Time of each Person $\rightarrow (2)$

In here all the times are represented in hours and we assumed all the other modules' capacity details are also the same as this value.

3. Order Time Calculations

Table 4.3: Order Time Calculation

Qty	Bulk SMV	Target Eff.	Time
5	15	6%	20.83

❖ *Bulk SMV* = Standard Minute value arrived at for a task based on the average rate of output while qualified workers will naturally achieve without over-exertion provided that they know and adhere to the specified method and that they are motivated to apply themselves to their work.

❖ *Target efficiency* = 6%

Because every work is done by one person they are set 6% as the target efficiency.

❖ *Oder Time* = $\frac{Qty * 60 * Bulk\ SMV}{Target\ Efficiency * 60}$ (in hours) $\rightarrow (3)$

According Table 5.3,

$$\text{Oder time} = \frac{5 * 15}{6\% * 60} = 20.833$$

Equation (3) Oder Time is get divided by 60 because we need to convert that time into the hours.

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Table 4.4: Time Schedule

Date	Start time	Break 1 Start	Break 1 End	Break 2 Start	Break 2 End	Break 3 Start	Break 3 End
Date	8:00	12:45	13:15	15:30	15:45	18:30	18:45

Table 4.4 shows their work starting time 8.00 a.m. and

Break 1 = Tea time

Break 2 = Lunch time

Break 3 = Evening tea time

05. RESULTS AND DISCUSSION

Mainly we use Transportation algorithm to solve the problem. In that case we want to get number of orders related to five style and total time duration related to that style as follows.

Table 5.1: Orders Group by Styles

Style Name	Number of Orders	Total Time(hours)
FLARE LEGGING	2	28.40
CS MODAL RIB HENLEY BODYSUIT	2	26.96
MESH RACERBACK FLOW DRESS	2	108.15
FITTED LONG SLEEVE CREW NECK MESH SHIRT	2	13.20
SWING LONG SLEEVE SCOOP NECK TUNIC	2	20.00

We get so many orders in different customers and we can group by five style. You can see what are the five styles by this table. In this case there are two orders to all style. Then we can get total time by number of pieces in an order multiply by time duration to one piece. All orders don't have same number of pieces. Finishing time duration to a one piece different by style to style.

We grouped orders by five styles then assign that styles to modules because we have five line then all styles can assign to correct lines as follows,

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Table 5.2: Final Result

	1	2	3	4	5
A	0	0	0	0	1
B	0	0	1	0	0
C	1	0	0	0	0
D	0	1	0	0	0
E	0	0	0	1	0

A - FLARE LEGGING

B – CS MODEL RIB HENLEY BODYDRESS

C – MESH RACERBACK FLOW DRESS

D – FITTED LONG SLEEVE CREW NECK MESH SHIRT

E – SWING LONG SLEEVE SCOOP NECK TUNIC

1 – First Module

2 – Second Module

3 – Third Module

4 – Fourth Module

5 – Fifth Module

In a sample room, there are five modules. These are denoted with numbers 1 to 5. To use the transportation algorithm in practically with our assumption we want five things. These five things are the styles. It's denoted with letters.

When we consider order by order we get less than five orders or more than five orders. In the less than five order case we can assign orders to modules but in the other case, we can't assign orders practically with assumption. Therefore we grouped orders by styles then we get five styles and we can assign those styles to modules. When one style is assigned to a correct line then all orders in each style are assigned to that module. After finishing the first order then start the second order in each style.

When all conditions are satisfied, then styles are assigned to the correct modules .it's denoted '1' in this table. Result is,

- FLARE LEGGING Style assigned to the Fifth module
- CS MODEL RIB HENLEY BODYDRESS Style assigned to the third module

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- MESH RACERBACK FLOW DRESS style assigned to the first module
- FITTED LONG SLEEVE CREW NECK MESH SHIRT style assigned to second module
- SWING LONG SLEEVE SCOOP NECK TUNIC style assigned to fourth module

Each styles have two orders. When we consider orders following table give how orders assign to modules

Table 5.3 : Orders vs. Modules

SRF ID	Modules
MRFD0001	Module 01
FLSC0001	Module 02
CMRH0001	Module 03
SLSS0001	Module 04
FLEG0001	Module 05
MRFD0002	Module 01
FLSC0002	Module 02
CMRH0002	Module 03
SLSS0002	Module 04
FLEG0002	Module 05

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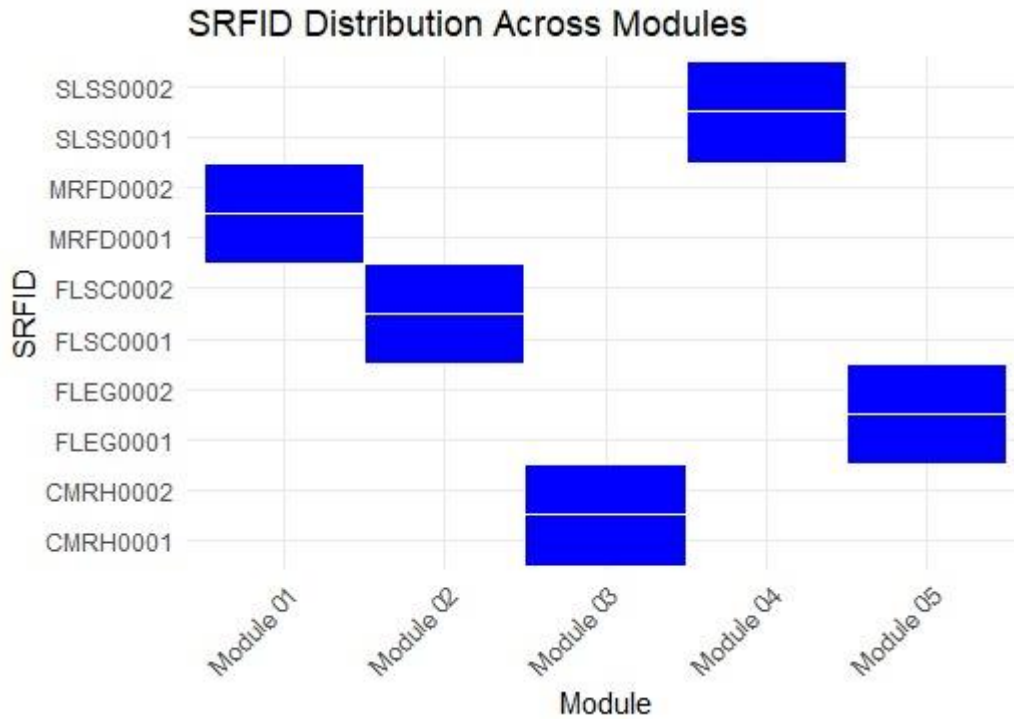


Figure 5.1: Order Assigning Gant Chart

All orders have a unique ID that is SRF ID. MRFD0001 and MRFD0002 orders in MESH RACERBACK FLOW DRESS style. These are assigned to the first module. FLSC0001 and FLSC0002 orders in FITTED LONG SLEEVE CREW NECK MESH SHIRT style. These are assigned to the second module. CMRH0001 and CMRH0002 orders in CS MODEL RIB HENLEY BODYSUIT Style. These are assigned to the third module. SLSS0001 and SLSS0002 orders in SWING LONG SLEEVE SCOOP NECK TUNIC style. These are assigned to the fourth module. FLEG0001 and FLEG0002 orders in FLARE LEGGING Style. These are assigned to the fifth module.

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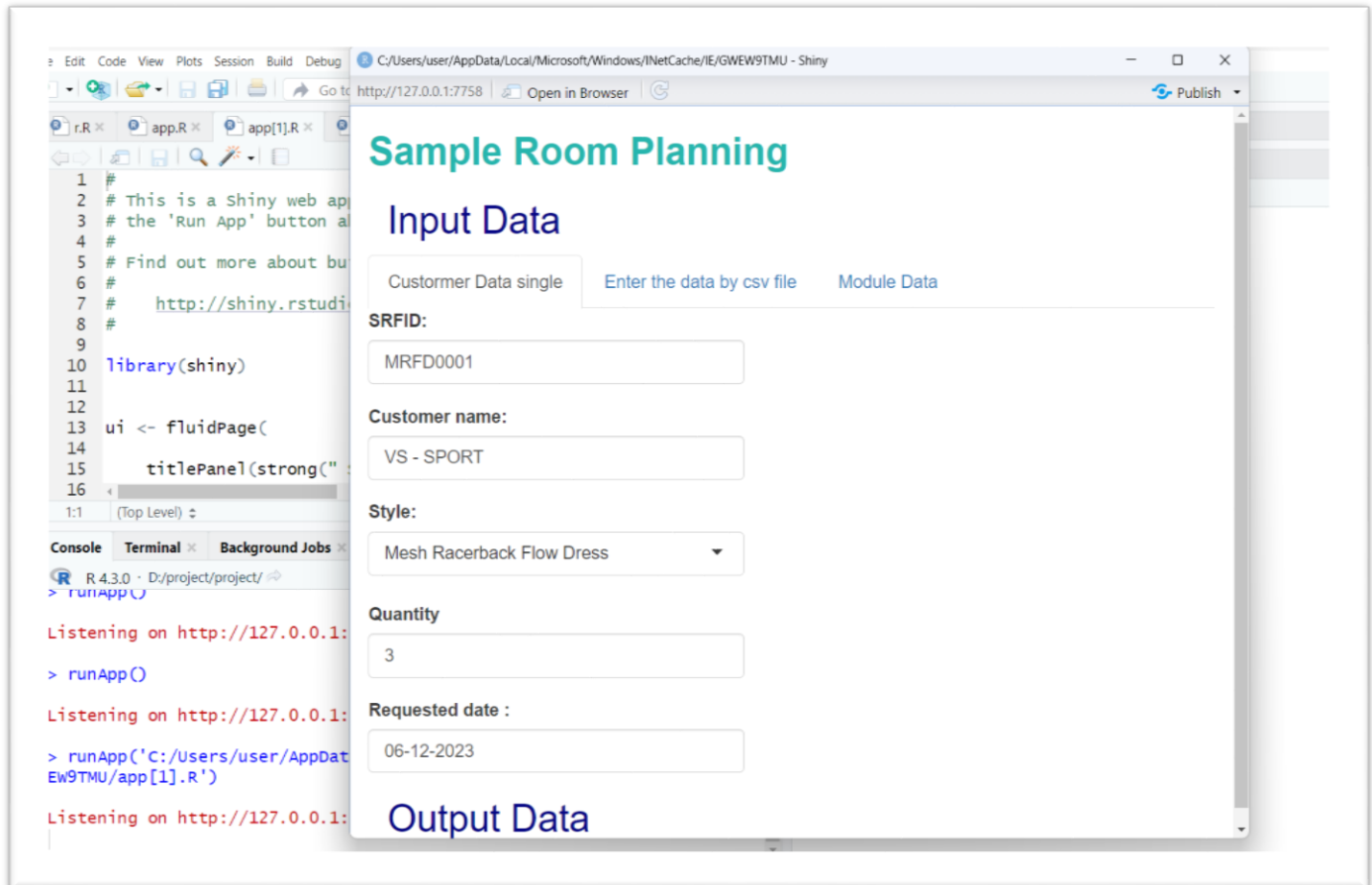


Figure 5.2 : Shiny Interface

We created a user-friendly user interface. It is implemented as a shiny app and it shows the final output.

We tried to use critical path method and job sequencing method but we faced a few problems so we moved to the transportation algorithm.

The Critical Path Method (CPM) is a simple but powerful technique for analyzing, planning, and scheduling large, complex projects. It is used to determine a project's critical path—the longest sequence of tasks that must be finished for the entire project to be completed. One project has more than one task but in our case, we have one task, which is the sewing task. Therefore we couldn't use critical path method for our problem.

In job sequencing method has several machines (modules) and one order goes from machine to machine. But in our case entire finishing duration of an order should be in one module. Then we couldn't use the job sequencing method.

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Acknowledging these drawbacks is essential for ongoing improvement and refinement of the optimization model, ensuring its relevance and effectiveness in real-world manufacturing scenarios.

- **Dependency on Input Accuracy:**

The accuracy of the optimization model heavily relies on the precision of input data, such as sewing times, demand forecasts, and machine capacities. Inaccuracies in these inputs may lead to suboptimal or unrealistic production plans.

- **Limited Consideration of External Factors:**

The project may have limitations in considering external factors, such as sudden machine breakdowns, unexpected delays in the supply chain, or changes in market demand. These uncertainties can impact the real-world applicability of the optimization model.

- **Assumption of Static Sewing Times:**

The project assumes static sewing times for each style, which may not accurately reflect the dynamic nature of manufacturing processes. Sewing times can vary based on factors like worker experience, material quality, and machine conditions.

- **Simplified Worker Skill Levels:**

The model assumes uniform skill levels among workers, overlooking potential variations in skill levels that could affect the efficiency of the sewing process. Incorporating a more nuanced understanding of worker skills could enhance the model's accuracy.

- **Lack of Integration with Human Factors:**

The project may not fully account for human factors, such as worker fatigue, motivation, or the impact of work breaks on overall productivity. Integrating these factors could provide a more realistic representation of the production environment.

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- **Single Objective Optimization:**

The project primarily focuses on minimizing production costs or maximizing resource utilization, potentially overlooking other critical objectives, such as worker satisfaction, which could affect long-term productivity and retention.

- **Difficulty in Modeling Complex Constraints:**

Certain constraints, such as intricate dependencies between different sewing styles or complex machine setup times, may be challenging to model accurately. This could lead to oversimplifications that impact the model's predictive capabilities.

- **Overlooking Urgency/Priority Orders:**

The model may not adequately address the prioritization of urgent orders with tight delivery deadlines. Failing to prioritize critical orders may result in customer dissatisfaction and missed business opportunities.

- **Inability to Address Seasonal Variations:**

The project may struggle to adapt to seasonal variations in demand, as it assumes a static representation of sewing times and capacities. Seasonal fluctuations could lead to suboptimal production plans during peak periods.

06. CONCLUSION

In conclusion, this project encompassed multiple intricacies within the sample room, with a primary focus on reducing planning efforts by 50%. Our overarching objective was to bolster efficiency through the elimination of manual data entry. Within the sample room, specialized modules dedicated to each style proved integral. Assigning sample orders to these modules emerged as a pivotal strategy, fostering precision and competitiveness.

During the implementation of our models, we confronted specific limitations, including the uniform capacity of all five modules, the potential for a single order to be assigned to multiple modules, and the simultaneous receipt of all orders. Additionally, customer exclusivity and style variations added complexity to the assignment process. Notably, the time required to sew samples remained consistent across sample types (proto, testing, try, photo shoot) and sizes (XS, S, and M).

Strategically employing the Transportation Algorithm played a key role in addressing the challenges associated with optimizing sample order assignments. This algorithm proved effective in navigating the constraints of equal module capacity and simultaneous order receipt.

In the final step, we determined the hours each module contributes to each style, with a focus on minimizing the weighted time. This crucial aspect of our approach ensures an optimized allocation of resources within the sample room, aligning seamlessly with our overarching goal of reducing planning efforts and enhancing overall operational efficiency.

06.01 Future works

We would like to enhance our code to integrate real-time dynamic data. Also incorporating machine learning techniques to predict the sewing times if new styles arrive. We also like to extend our model to address multiple objectives. Also, we are planning to develop a more user-friendly user interface that is easy to interact with all the decision-makers in the sample room. Finally, we would like to compare our solution with industry standards and identify advanced improvements that can be made.

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